

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2025

Course Code: MAT102R01

Course: ENGINEERING MATHEMATICS - II

QP No. :U052-2

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. Find the radius of curvature of $y = e^x$ at the point where it crosses the y axis.
2. Find the envelope of family of curves $x \cos \alpha + y \sin \alpha = a \sec \alpha$, where α being the parameter.
3. Examine the nature of stationary point $(5, -1)$ of the function $f(x, y)$ if, $f_{xx} = 6x - 30$, $f_{xy} = 6y$, $f_{yy} = 6x - 30$.
4. In a culture of yeast, if the active ferment doubles itself in 3 hours, by what ratio will it increase in 15 hours on the assumption that the quantity increase at a rate proportional to itself?
5. Find the particular integral of $(D^2 - 4D + 4)y = 3^x$.
6. Determine the value of a for which the following differential equation is exact: $(2xy + 3y^2)dx + (ax^2 + 6xy)dy = 0$.
7. If $\phi = x^3 + y^3 + z^3 - 3xyz$, find $\nabla\phi$ at $(1, 2, 3)$.
8. If $\vec{A}$ and $\vec{B}$ are irrotational, prove that $\vec{A} \times \vec{B}$ is solenoidal.

9. State Gauss divergence theorem.

10. Prove that the area bounded by a simple closed curve C is given by $\frac{1}{2} \int_C (x dy - y dx)$.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Find the radius of curvature at the point (r, θ) on the curve $r^2 \cos 2\theta = a^2$. (8)

(b) Discuss the maxima and minima of the function $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$. (7)

(OR)

12. (a) Find the envelope of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where the parameters a and b are connected by the relation $a^n + b^n = c^n$, c is a constant. (8)

(b) The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$. (7)

13. (a) Solve $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$. (8)

(b) A Voltage Ee^{-at} is applied at $t = 0$ in a circuit containing inductance L and resistance R in series. Show that the current i at any time t is $\frac{E}{R-aL} \left(e^{-at} - e^{-\left(\frac{R}{L}\right)t} \right)$, assuming that $i = 0$ at $t = 0$. (7)

(OR)

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div } \vec{F} dv$$

14. (a) A cup of coffee at temperature 90°C is placed in a room with temperature as 25°C and it cools to 60°C in 5 minutes. Find its temperature after a further interval of 5 minutes. Also, find the time at which the temperature of coffee will come down further by 20°C . (8)

(b) Solve $(D^2 + 1)y = \operatorname{cosec} x \cot x$ using method of variation of parameters. (7)

15. (a) Find the angle between the surfaces $x^2 - y^2 - z^2 = 11$ and $xy + yz - zx = 18$ at the point $(6, 4, 3)$. (8)

(b) Find the directional derivative of the function $\phi = xy^2 + yz^2$ at the point $(2, -1, 1)$ in the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at the point $(-1, 2, 1)$. (7)

(OR)

16. (a) If $r = |\vec{r}|$, where $\vec{r}$ is the position vector of the point (x, y, z) with respect to the origin, prove that (i) $\nabla f(r) = \frac{f'(r)}{r} \vec{r}$ and $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$. (8)

(b) Find the values of a, b, c so that $\vec{F} = (x + 2yaz)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational. (7)

17. Verify Green's theorem in a plane for $\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region is defined by the lines $x = 0, y = 0, x + y = 1$.

(OR)

18. Verify Stoke's theorem for $\vec{F} = -y\vec{i} + 2yz\vec{j} + y^2\vec{k}$, where S is the upper half of the sphere $x^2 + y^2 + z^2 = a^2$ and C is the circular boundary on the xoy plane.

PART - C

Answer the following

1 x 20 = 20 Marks

19. (a) Find the evolute of the rectangular hyperbola $xy = c^2$. (10)

- (b) Solve $(x^2 D^2 + xD + 1)y = \log x \sin(\log x)$. (10)
